(which will in a very short time undergo flavour oscillations) with a probability of about 99.99%. The second most probable fate of the pion is by decaying into an electron and a electron neutrino with a probability of about 0.01%. This latter decay was discovered in CERN in 1958. If none of the previous decay modes are realised they collide with other particles in the atmosphere and are annihilated. The muon is also an unstable subatomic particle with a mean lifetime for stationary muons of 2.2 μs. This comparatively long decay life time (the second longest known) is due to being mediated by the weak nuclear force. The only longer lifetime for an unstable subatomic particle is that for the free neutron which also decays via the weak force. All muons decay to three particles: an electron plus two neutrinos of different types.

The collision reactions of the pions result in the formation of a large number of other particles such as electrons, neutrons, protons, and photons. Some of the electrons so formed are captured in a thick zone around the earth known as the inner van Allen belt.

The main part (50–80%) of the cosmic radiation which reaches the earth's surface consists of high energy muons. Muons have much less tendency to react with atomic nuclei than pions and, therefore, can penetrate the atmosphere and the solid material of the earth relatively easily. The remaining part, which is the lower energy component of the cosmic radiation that strikes the earth, consists of photons, electrons, and positrons. At sea level this part of the cosmic radiation gives rise to approximately 2–3 ion pairs s^{-1} cm^{-3} of air. It is this component of the cosmic radiation that gives rise to the cosmic ray portion of the natural background that is measured by nuclear detection devices in laboratories.[1] Some of the cosmic radiation interacts to make atmospheric radioactivity which is principally the nuclides ^{3}H and ^{14}C.

13.3. RADIOELEMENTS IN NATURE

Practically all natural materials contain radioactive nuclides, though usually in such low concentrations that they can be detected only by very sensitive analysis. This is true for water (rain water, rivers, lakes, sea), rocks and soil, and all living matter, as well as for structures based on natural raw materials except where great care has resulted in use of a non-radioactive material. The radionuclides in our environment can be divided into (i) those formed from cosmic radiation, *cosmogenic*, (ii) those with lifetimes comparable to the age of the earth together with those that are part of the natural decay chains beginning with thorium and uranium, *primordial*, and (iv) those introduced into nature by man using modern techniques, *anthropogenic*.

[1] The remainder of the natural background comes from naturally occurring radioactive elements in the laboratory materials and surrounding building.

13.4. COSMOGENIC RADIONUCLIDES

13.4.1. Survey

Cosmic irradiation of the atmosphere produces neutrons and protons which react with N_2, O_2, Ar, etc. resulting in the production of radioactive nuclides, some of which are listed in Tables 13.1 and 13.2. These nuclides are produced at constant rates and brought to the earth surface by rain water. Though they are formed in extremely low concentrations, the global inventory is by no means small (§§13.4.2 and 13.4.3). Equilibrium is assumed to be established between the production rate and the mean residence time of these radionuclides in terrestrial reservoirs (the atmosphere, the sea, lakes, soil, plants, etc) leading to constant specific radioactivity of the elements in each reservoir. If a reservoir is closed from the environment, its specific radioactivity decreases. This can be used to determine exposure times of meteorites to cosmic radiation (and the constancy of

Table 13.1 Long-lived cosmogenic radionuclides appearing in meteorites and rain water

Nuclide	Half-life (years)	Decay mode & particle energy (MeV)	Atmospheric production rate (atoms m^{-2} s^{-1})
^{3}H	12.32	β^- 0.0186	2500
^{10}Be	1.52×10^6	β^- 0.555	300
^{14}C	5715	β^- 0.1565	17 000—25 000
^{22}Na	2.605	β^+ 0.545	0.5
^{26}Al	7.1×10^5	β^+ 1.16	1.2
^{32}Si	160	β^- 0.213	1.6
^{35}S	0.239 (87.2 d)	β^- 0.167	14
^{36}Cl	3.01×10^5	β^- 0.709	60
^{39}Ar	268	β^- 0.565	56
^{53}Mn	3.7×10^6	EC (0.596)	
^{81}Kr	2.2×10^5	EC (0.28)	

Values within parenthesis after EC are decay energies.

Table 13.2 Short-lived cosmogenic radionuclides appearing in rain water

	Decay mode	
Nuclide	Half-life	Particle energy (MeV)
^{7}Be	53.28 d	EC (0.862)
^{24}Na	14.96 h	1.389
^{28}Mg	21.0 h	0.459
^{32}P	14.28 d	1.710
^{33}P	25.3 d	0.249
^{39}Cl	55.6 min	1.91

Value within parenthesis after EC is decay energy
Production rates (atoms m^2 s^1): ^{7}Be 81, ^{39}Cl 16